

function of the car on the repair bench itself before leaving it to move on the ground. You can lift the car, switch on the power and check the operation by putting your hand near the sensor to ensure each sensor is working perfect-

ly. You can adjust the range of sensing by varying the preset VR1. It is not recommended to keep the range very long because this would cause the car to keep moving forward and backward as it senses any obstacle, even far away

from it. Once you are sure that the car is operating well, you can leave it to move freely.

EFY note. The source code of this article is included in this month's EFY-CD.

ROBO1.C

```
#include <at89x52.h>
void delay_ret(void);
void delay_step(void);
void left(void);
void right(void);

int i,j,k;
void main(void)
{
    P0=0xFF; /*initialise P0 as input
    for sensor*/
    for (i=0;i<10000;i++);

    while(1)
    {
        P3=0xF2; /*move forward*/

        switch(P0) /*check status of
        port P1*/
        {
            case(0xFB):
                P3=0xF1; /*apply reverse direction
                to stop car*/
                for (i=0;i<30000;i++);
                P3=0x00; /*stop
                car*/
                right(); /*turn
                right*/
                P1=0;
                P3=0xF1; /*move reverse*/
                delay_ret(); /*delay*/
                delay_ret(); /*delay*/
                left(); /*return to center*/
                P1=0;
                P3=0x00; /*stop the car*/
                for (i=0;i<3000;i++); /*wait a while*/
                P3=0xF2; /*move forward*/
                break;

            case(0xFE):
                P3=0xF1; /*apply reverse direction
                to stop car*/
                delay_ret();
                P3=0x00; /*stop
                car*/
                left(); /*turn
                left*/
                P1=0;
                P3=0xF2; /*move ahead*/
                delay_ret(); /*wait for some time*/
                right(); /*return to center posi-
                tion*/
                P1=0;
                break;

            case(0xFD):
                P3=0xF1; /*apply reverse direction
                to stop car*/
                delay_ret();
                P3=0x00; /*stop
                car*/
                right(); /*turn
                right*/
                P1=0;
                P3=0xF2; /*move ahead*/
                delay_ret(); /*wait for some time*/
                left(); /*return to center posi-
                tion*/
                P1=0;
                break;

            }

            void delay_ret(void) /*delay
            routine to wait before*/
            {
                /*steer-
                ing is returned back to straight*/
                int x,t;
                for (t=0;t<600;t++)
                {
                    for (x=0;x<400;x++);
                }
            }

            void delay_step(void) /*delay
            routine for controlling speed of step-
            per*/
            {
                int j;
                for (j=0;j<500;j++)
                {
                    void left(void) /*turn
                    steering left*/
                    {
                        for (k=0;k<3;k++)
                        {
                            P1=0x05;
                            delay_step();
                            P1=0x09;
                            delay_step();
                            P1=0x0A;
                            delay_step();
                            P1=0x06;
                            delay_step();
                            P1=0x00;
                        }
                    }

                    void right(void) /*turn
                    steering right*/
                    {
                        for (k=0;k<3;k++)
                        {
                            P1=0x05;
                            delay_step();
                            P1=0x06;
                            delay_step();
                            P1=0x0A;
                            delay_step();
                            P1=0x09;
                            delay_step();
                            P1=0x00;
                        }
                    }
                }
            }
        }
    }
}
```

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